Part B:Maximizing Revenue – POS Data Analysis for Combo Offers in Grocery Stores

Name: Sattaru Manohar
Batch : PGPDSBA.O.APR 23.A
Module: Marketing and Business Analysis

Executive Summary

- The analysis of transactional data from the grocery store's Point of Sale system with 20,641 entries and 3 columns identified notable trends, including a peak in orders in 2018 and varying quarterly order patterns.
- Top 10 items, such as Poultry, Soda, and Cereals, were identified, with Order ID 226 having the highest number of items (34). Weekly data revealed a slight downward trend in orders week on week.
- Market Basket Analysis with KNIME established association rules based on a minimum support of 0.042 and a confidence of 0.5, ensuring the consideration of frequent associations and reliable rules.
- **Recommendations for enticing combo offers include:**

  Combo 1: Yogurt, Cheeses, Coffee/Tea, Pork
- Offer: Buy all items together and get a discount.
  Combo 2: Cheeses, Coffee/Tea, Pork
- Offer: Buy these items together for a special price.
  Combo 3: Yogurt, Waffles, Soda, Sandwich Bags
- Offer: Special discount for purchasing these items together.

The rationale behind these offers is the frequent co-occurrence of items in transactions, aiming to encourage customers to buy more through attractive combo deals.

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Problem Statement

- Analyze transactional data from a grocery store's Point of Sale system.
- Identify frequently co-purchased items to enhance revenue.
- Provide strategic recommendations for enticing combo offers and discounts.
- Conduct exploratory analysis of data trends across months, years, quarters, and days.
- Utilize Market Basket Analysis with KNIME to establish association rules.
- Create a KNIME workflow image and discuss threshold values for Support and Confidence.

Summary of the data

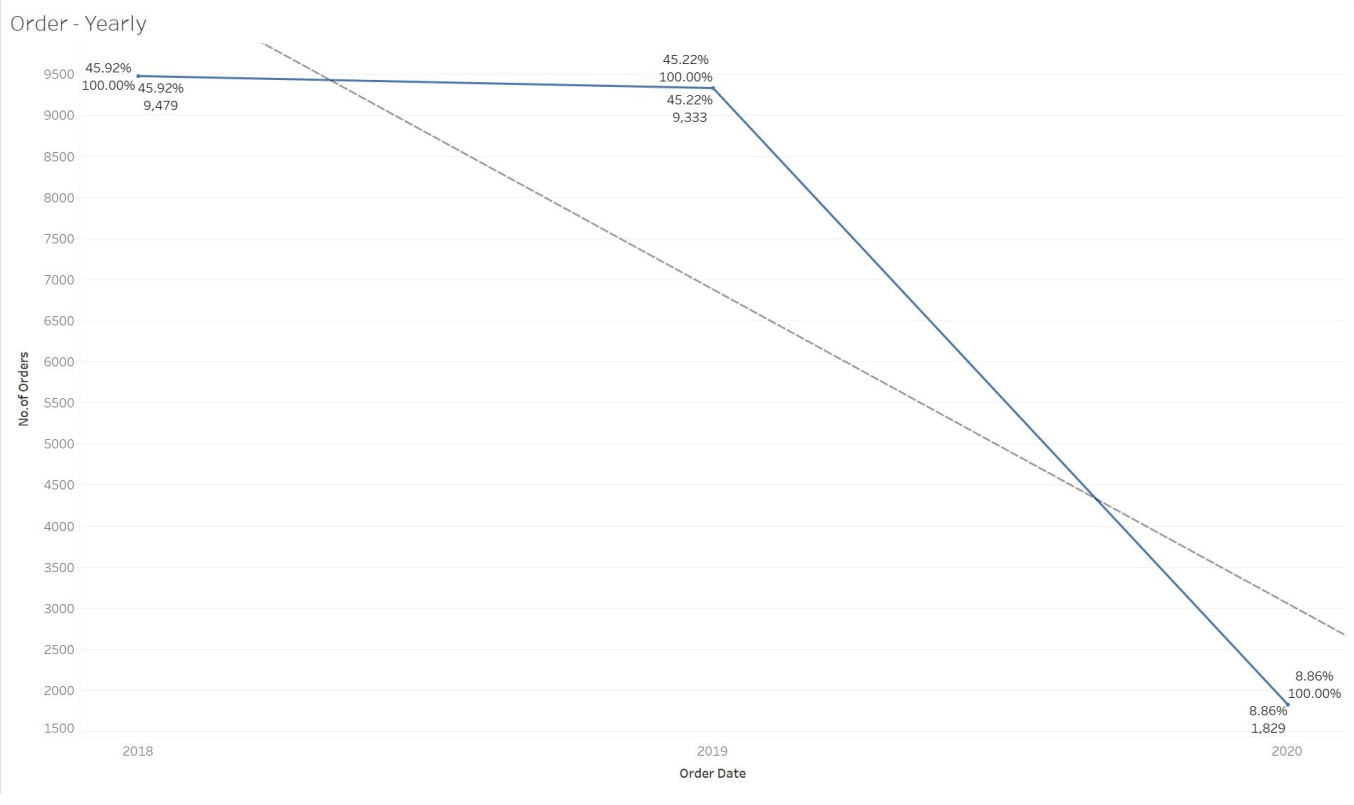
	Order_id
count	20641.000000
mean	575.986289
std	328.557078
min	1.000000
25%	292.000000
50%	581.000000
75%	862.000000

- The dataset comprises **20641** entries with **3** columns, including order details, date and Item name.
- No Null values found in the data set

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20641 entries, 0 to 20640
Data columns (total 3 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Date       20641 non-null  object
 1   Order_id   20641 non-null  int64
 2   Product    20641 non-null  object
```

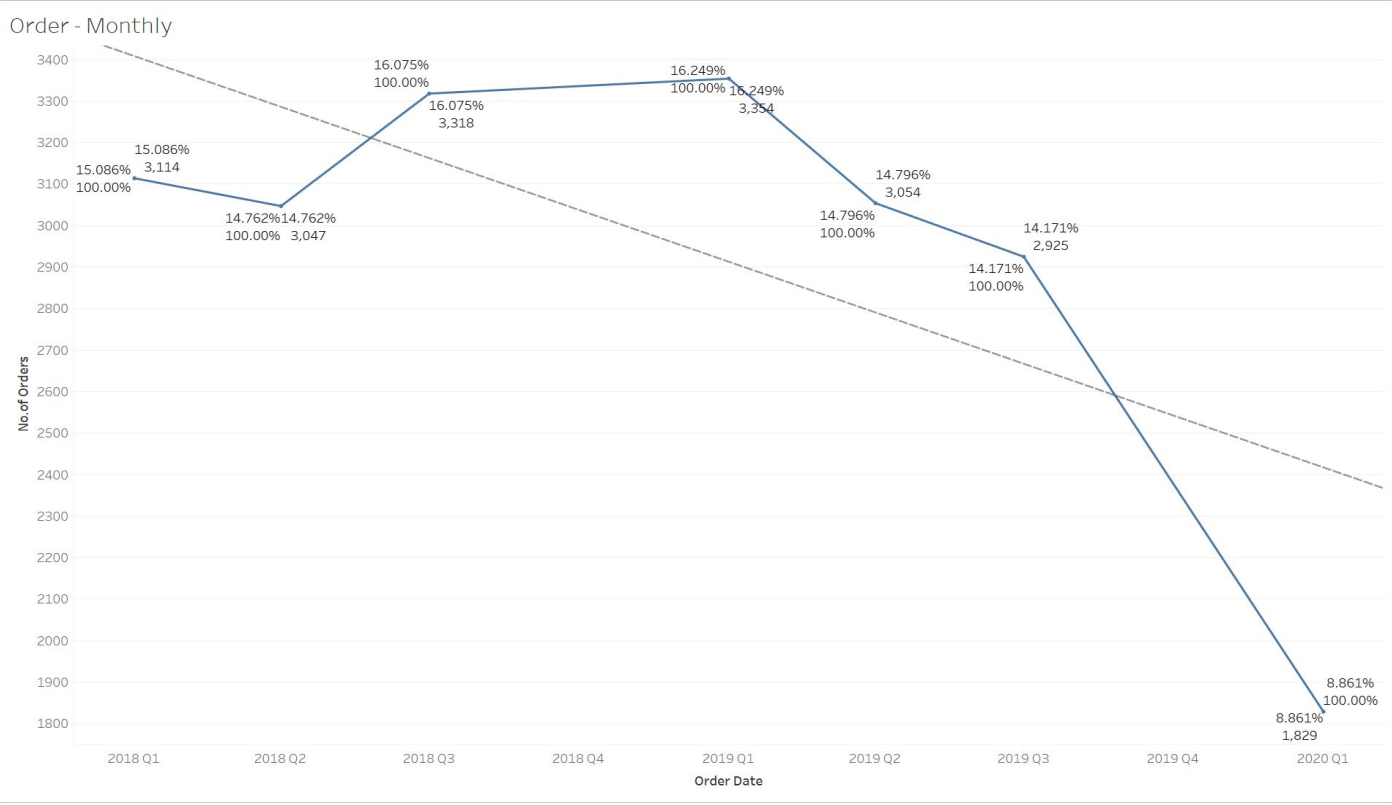
Exploratory Data Analysis – Grocery Stores

- 2018 has the highest number of orders 9479



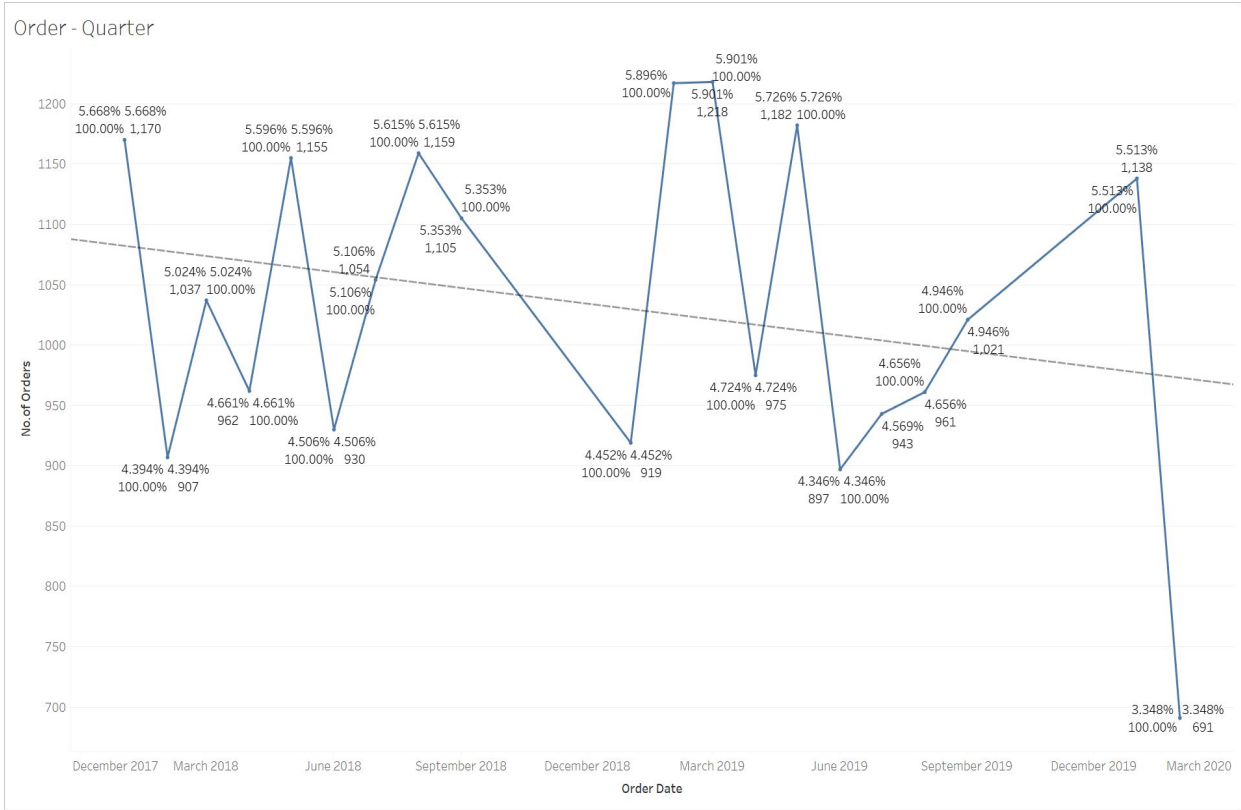
Exploratory Data Analysis – Grocery Stores

- As per the data the orders are eventually decreased as shown in the trend line from 2018 to 2020



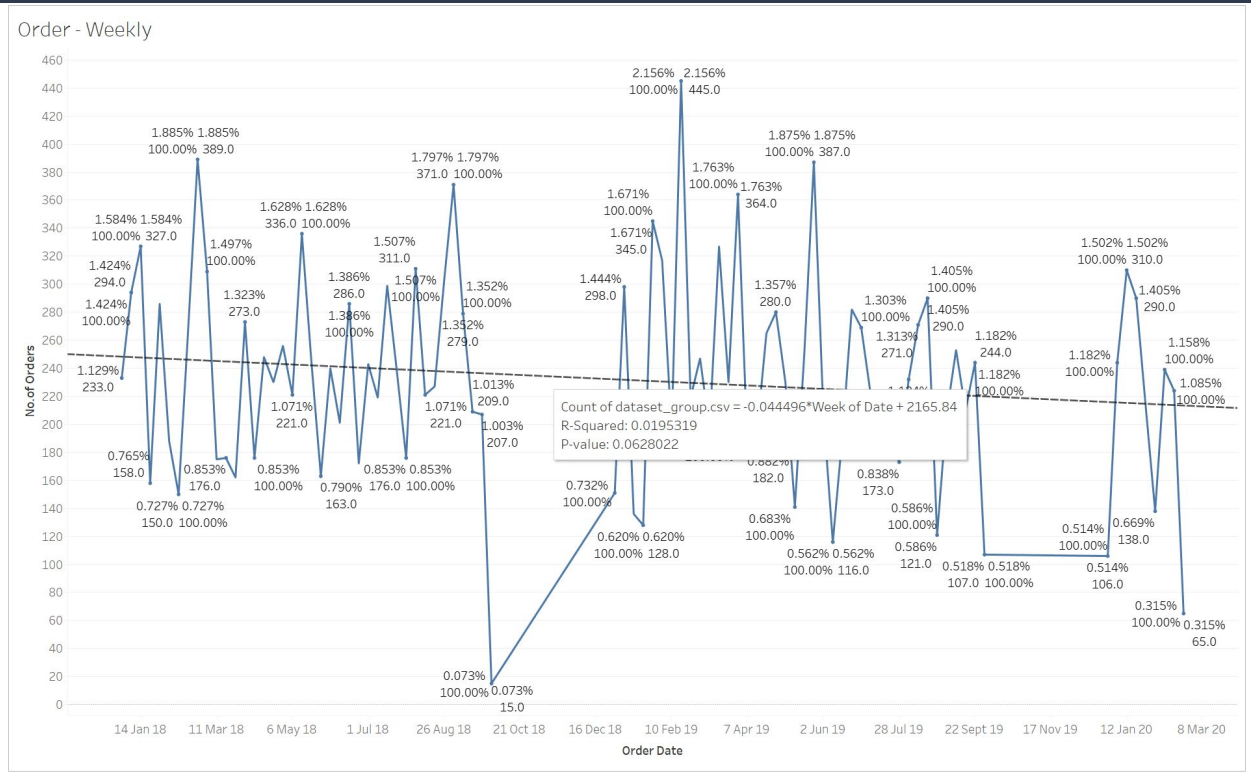
Exploratory Data Analysis – Grocery Stores

- As per the data the orders are decreased in some quarters and Increased in some other quarters



Exploratory Data Analysis – Grocery Stores

- As per the data, when it comes to weekly data the the orders has slightly down trend week on week.

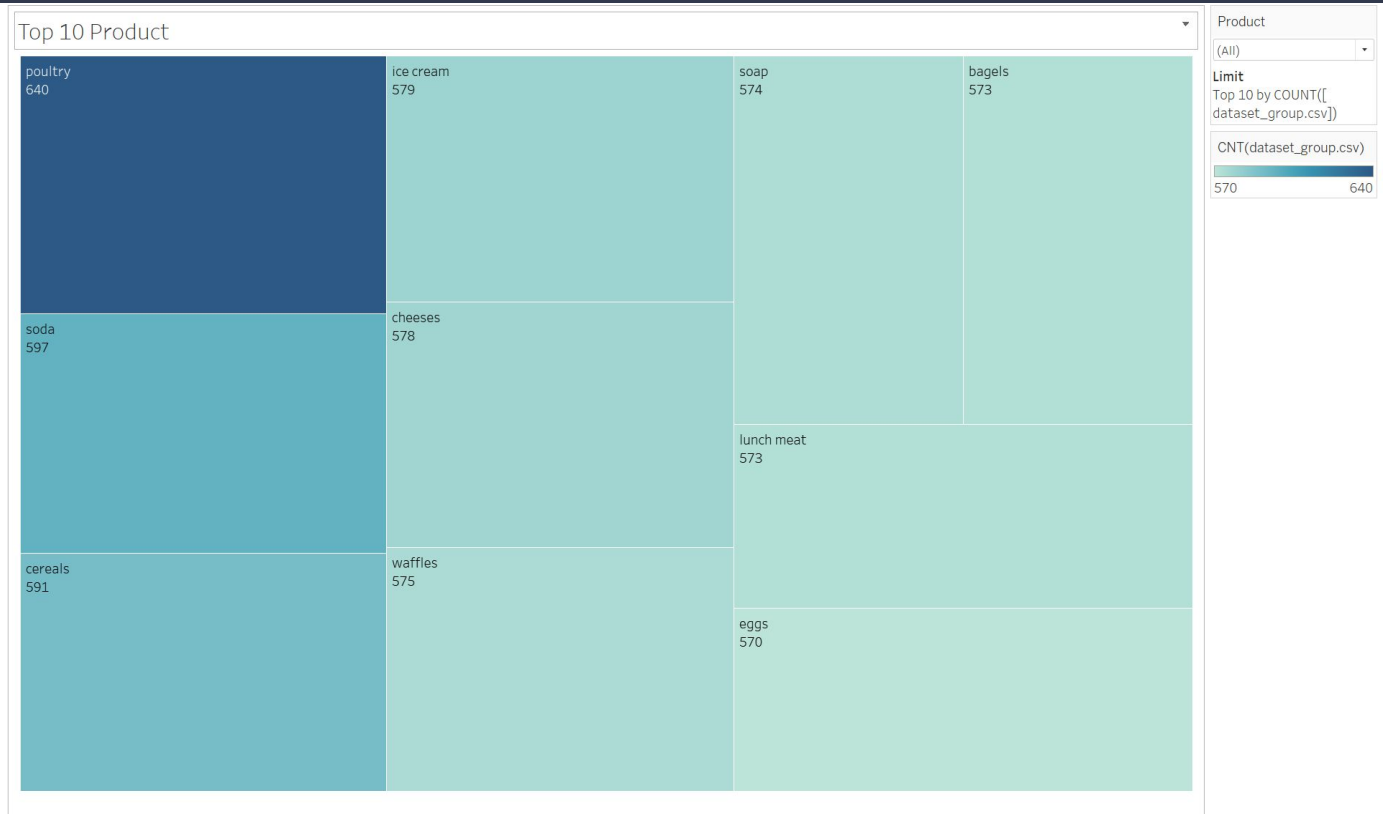


Exploratory Data Analysis – Grocery Stores

- List of items represented in in the world cloud

Exploratory Data Analysis - Grocery Stores

- Top 10 Items sold are mentioned
- Poultry - **640**
- Soda - **597**
- Cereals - **591**



Exploratory Data Analysis – Grocery Stores

- Monthly wise product list and product count count

Product wise order - Monthly

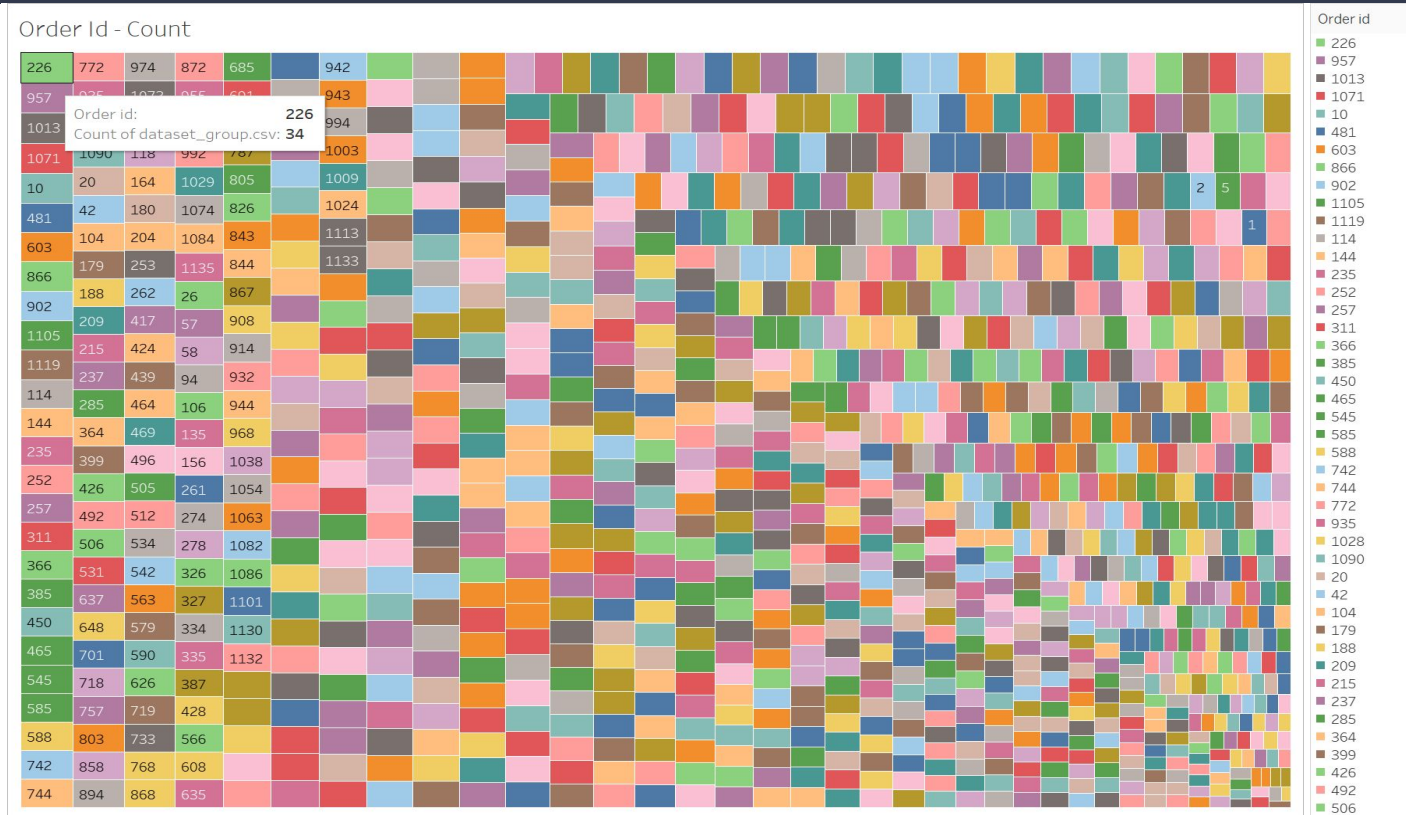
Product	January	February	March	April	May	June	July	August	September
poultry	94	94	70	65	66	55	63	60	73
soda	97	79	56	55	72	48	52	69	69
cereals	95	82	57	52	72	49	52	74	58
ice cream	95	83	59	61	47	53	60	67	54
cheeses	91	87	67	50	44	65	56	58	60
waffles	91	82	55	48	71	46	53	73	56
soap	85	70	70	55	68	41	55	69	61
lunch meat	78	86	51	52	57	51	79	59	60
bagels	94	80	73	60	56	56	40	64	50
juice	91	66	74	60	70	36	48	71	54
eggs	86	67	78	59	70	47	49	51	63
toilet paper	86	79	63	54	78	54	49	51	55
dinner rolls	94	81	60	67	60	41	65	53	46
aluminum foil	85	70	59	62	53	56	55	59	67
coffee/tea	94	60	60	50	59	64	68	52	58
shampoo	97	74	64	44	51	55	61	60	56
beef	87	89	67	43	64	58	45	57	51
paper towels	92	77	60	54	75	46	53	41	58
milk	84	81	53	54	67	48	57	56	55
flour	89	65	57	55	62	49	69	49	60
butter	86	64	73	43	83	48	42	49	67
mixes	90	93	59	42	52	49	57	55	57
dishwashing liqu..	90	87	69	52	60	41	49	51	52
all- purpose	91	77	58	55	62	46	48	53	61
ketchup	91	80	59	43	70	42	52	51	60
yogurt	79	70	51	60	76	46	50	59	54
individual meals	85	65	61	58	51	47	46	58	73
tortillas	85	65	53	47	76	52	56	55	54
pasta	83	67	50	66	65	51	60	47	53
laundry detergent	82	67	60	41	61	57	43	69	62
spaghetti sauce	73	65	73	60	61	47	52	57	48
sandwich bags	67	82	66	46	67	51	48	53	56
sugar	75	79	54	40	56	53	59	51	66
pork	92	77	60	44	48	44	54	50	62
fruits	72	75	55	48	70	50	44	72	43
sandwich loaves	99	69	51	48	61	45	58	51	41
hand soap	82	81	50	44	56	40	50	46	53

Product
- all- purpose
- aluminum foil
- bagels
- beef
- butter
- cereals
- cheeses
- coffee/tea
- dinner rolls
- dishwashing liquid/de..
- eggs
- flour
- fruits
- hand soap
- ice cream
- individual meals
- juice
- ketchup
- laundry detergent
- lunch meat
- milk
- mixes
- paper towels
- pasta
- pork
- poultry
- sandwich bags
- sandwich loaves
- shampoo
- soap
- soda
- spaghetti sauce
- sugar
- toilet paper
- tortillas
- waffles
- yogurt

CNT(dataset_group.csv)
- 36
- 60
- 80
- 99

Exploratory Data Analysis – Grocery Stores

- Order ID 226 has the highest number of Items i.e 34 Items



Application of Market Association Rules

Knime Flowchart diagram:

Minimum Support (0.042) which means that only associations with itemsets occurring in at least 4.2% of all transactions are considered. This helps filter out less common associations, focusing on those with a certain level of frequency.

Minimum Confidence (0.5): A confidence of 0.5 indicates that the rule is considered significant if the likelihood of the consequent given the antecedent is at least 50%. Rules with a confidence below this threshold are typically considered less reliable.

Interpretation: A rule with a minimum support of 0.042 ensures that only associations frequent enough in the dataset are considered.

A confidence of 0.5 ensures that only rules with a substantial likelihood of the consequent given the antecedent are considered.

Rule Outputs : 5422

#	RowID	Support Number (dou...	Confidence Number (dou...	Lift Number (dou...	Rec_Prod... String	implies String	Items Set	Basket Ite... String
1	rule0...	0.042	0.565	1.468	yogurt	<---	[cheeses,coff...	cheeses
2	rule0...	0.042	0.565	1.468	yogurt	<---	[cheeses,coff...	coffee/tea
3	rule0...	0.042	0.565	1.468	yogurt	<---	[cheeses,coff...	pork
4	rule1...	0.042	0.615	1.575	cheeses	<---	[yogurt,coffee...	yogurt
5	rule1...	0.042	0.615	1.575	cheeses	<---	[yogurt,coffee...	coffee/tea
6	rule1...	0.042	0.615	1.575	cheeses	<---	[yogurt,coffee...	pork
7	rule2...	0.042	0.593	1.562	coffee/tea	<---	[yogurt,chees...	yogurt
8	rule2...	0.042	0.593	1.562	coffee/tea	<---	[yogurt,chees...	cheeses
9	rule2...	0.042	0.593	1.562	coffee/tea	<---	[yogurt,chees...	pork
10	rule3...	0.042	0.545	1.418	yogurt	<---	[waffles,soda,...	waffles
11	rule3...	0.042	0.545	1.418	yogurt	<---	[waffles,soda,...	soda
12	rule3...	0.042	0.545	1.418	yogurt	<---	[waffles,soda,...	sandwich bags
13	rule4...	0.042	0.516	1.403	sandwich bags	<---	[yogurt,waffle...	yogurt
14	rule4...	0.042	0.516	1.403	sandwich bags	<---	[yogurt,waffle...	waffles

Based on the association rules generated from the market basket analysis, here are suggestions for possible combos with lucrative offers:

Combo 1: Yogurt, Cheeses, Coffee/Tea, Pork

- Offer: Buy all items together and get a discount.
- Rationale: These items frequently appear together in transactions, and offering a discount for purchasing the combo may encourage customers to buy more.

Combo 2: Cheeses, Coffee/Tea, Pork

- Offer: Buy these items together for a special price.
- Rationale: These items show a strong association, and creating a combo offer could increase sales for these specific products.

Combo 3: Yogurt, Waffles, Soda, Sandwich Bags

- Offer: Special discount for purchasing these items together.
- Rationale: These items are frequently bought in combination; providing a discount may attract customers looking for a bundled deal.

Based on the association rules generated from the market basket analysis, here are suggestions for possible combos with lucrative offers:

Combo 4: Yogurt, Milk, Mixes, Sandwich Bags

- Offer: Buy this combo and get a discount or a free item.
- Rationale: The association rules suggest that customers tend to purchase these items together, making it a potential combo for a special offer.

Combo 5: Waffles, Soda, Sandwich Bags

- Offer: Bundle these items for a discounted price.
- Rationale: These products are often bought together; offering them as a combo can enhance the perceived value for customers.

Combo 6: Yogurt, Cheeses, Pork

- Offer: Special price for purchasing these items together.
- Rationale: Capitalize on the association between these items to encourage customers to buy them in a single transaction.

Combo 7: Yogurt, Waffles, Soda

Based on the association rules generated from the market basket analysis, here are suggestions for possible combos with lucrative offers:

Combo 7: Yogurt, Waffles, Soda

- Offer: Combo deal with a discounted price.
- Rationale: These items have a strong association, and providing a bundled offer can attract customers looking for convenient combos.

Combo 8: Yogurt, Milk, Mixes

- Offer: Purchase this combo and receive a discount.
- Rationale: The association rules indicate a connection between these items, making them a suitable candidate for a combo offer.

These recommendations leverage the insights gained from market basket analysis, aiming to increase revenue by encouraging customers to purchase specific combinations of items through attractive and cost-effective offers.